

Proof. Step1. Indicator of the tail event. Define the indicator random variable

$$I(\omega) = \mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases}$$

By definition of expectation of an indicator,

$$\mathbb{E}[I] = \mathbb{P}(X \geq a).$$

Step2. Pointwise bound. For every $\omega \in \Omega$ we compare $I(\omega)$ with $X(\omega)/a$.

- If $X(\omega) < a$, then $I(\omega) = 0$ and $X(\omega)/a \geq 0$, so $0 = I(\omega) \leq X(\omega)/a$.
- If $X(\omega) \geq a$, then $I(\omega) = 1$ and $X(\omega)/a \geq a/a = 1$, hence $1 = I(\omega) \leq X(\omega)/a$.

Thus

$$I(\omega) \leq \frac{X(\omega)}{a} \quad \text{for all } \omega \in \Omega. \quad (1)$$

Step3. Take expectations (monotonicity). Both sides of (??) are non-negative random variables, so by monotonicity of the expectation operator,

$$\mathbb{E}[I] \leq \mathbb{E}\left[\frac{X}{a}\right] = \frac{1}{a} \mathbb{E}[X].$$

Recalling that $\mathbb{E}[I] = \mathbb{P}(X \geq a)$ we obtain

$$\boxed{\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}}.$$

Step4. Degenerate / edge cases.

- **Case $a \leq 0$.** The hypothesis of the inequality requires $a > 0$. If one allows $a \leq 0$, then $\{X \geq a\} = \Omega$ (since $X \geq 0$), so $\mathbb{P}(X \geq a) = 1$, while the right-hand side $\mathbb{E}[X]/a$ is either undefined (division by zero) or non-positive; the statement is then vacuous or interpreted as $1 \leq +\infty$.
- **Case $\mathbb{E}[X] = +\infty$.** The right-hand side is $+\infty$, which trivially dominates any probability, so the inequality holds.
- **Case $X \equiv 0$.** Both sides equal 0, giving equality.

In what follows we assume $a > 0$ and $\mathbb{E}[X]$ finite (the argument is unchanged when $\mathbb{E}[X] = +\infty$).

Step5. Equality condition. Equality in the chain

$$\mathbb{P}(X \geq a) = \mathbb{E}[I] \leq \mathbb{E}\left[\frac{X}{a}\right] = \frac{\mathbb{E}[X]}{a}$$

occurs iff equality holds in the pointwise bound (??) $\mathbb{P}$ -a.s., because expectation is strictly monotone: if $Y \leq Z$ a.s. and $\mathbb{E}[Y] = \mathbb{E}[Z]$, then $Y = Z$ a.s. Thus we require

$$I(\omega) = \frac{X(\omega)}{a} \quad \text{for } \mathbb{P}\text{-a.e. } \omega.$$

On $\{X < a\}$ we have $I = 0$, so the equality forces $X(\omega) = 0$ a.e. on this set. On $\{X \geq a\}$ we have $I = 1$, so we must have $X(\omega) = a$ a.e. on this set. Consequently,

$$X = a \mathbf{1}_{\{X \geq a\}} \quad \text{a.s.,}$$

i.e. X takes only the values 0 and a (the latter possibly with probability 0, which reduces to the trivial case $X \equiv 0$). This is exactly the condition for equality in Markov's inequality.

Step6. Conclusion. We have proved Markov's inequality for any non-negative random variable X and any real $a > 0$, treated the degenerate situations, and described precisely when equality occurs. □ □